Unit 2, Lesson 2: Types of Reasoning

Benchmark

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

Learning Targets

- ☐ I can write a conjecture.
- ☐ I can write a counterexample.
- ☐ I can use inductive and deductive reasoning.

2.2.1 Warm-Up: What's the Pattern?

Four steps of a pattern are shown.

Step 1	Step 2
	
Step 3	**Step 4**
	

1. How many blue circles will be in step 5?

2. How many orange circles will be in step 5?

3. How many green circles will be in step 5?

4. Describe the pattern using words or an algebraic expression.

2.2.2 Exploration: Types of Reasoning

To describe a pattern such as the pattern in the Warm-Up is to make a conjecture.

A ***conjecture*** is a mathematical statement that has not yet been proven.

Once a conjecture has been made, then reasoning can be used to prove the conjecture.

1. Work with your partner to write conjectures for the pattern in the Warm-Up. You should be able to use each conjecture to determine any step in the pattern.

   - number of blue circles
   - number of orange circles
   - number of green circles
   - total number of circles

Blue	Orange	Green	Total

A conjecture is an important step in problem-solving. Conjectures arise when you notice a pattern and that pattern holds true for many cases. However, just because a pattern holds true for many cases does not mean that the pattern will hold true for ***all*** cases. Conjectures must be proven for the mathematical observation to be fully accepted. When a conjecture is rigorously proven, it becomes a theorem.

The values for step 10 are shown. Use these values to test your conjecture. Revise your conjecture if necessary.

Blue	Orange	Green	Total
63	81	40	184

2. Ask a classmate for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the conjecture and summary columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Color	Conjecture	My Summary of Their Reason	Initials
Blue			
Orange			
Green			
Total			

The conjectures from your classmate may be different from your conjectures, but that does not mean that one is wrong. Both may be correct.

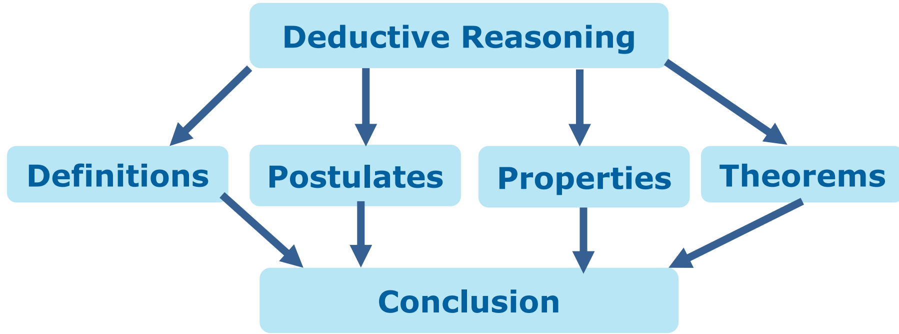
3. Compare your conjecture with the conjectures you and your partner collected from different classmates.

4. The values for step 18 are shown. Use these values to test your classmates' conjectures.

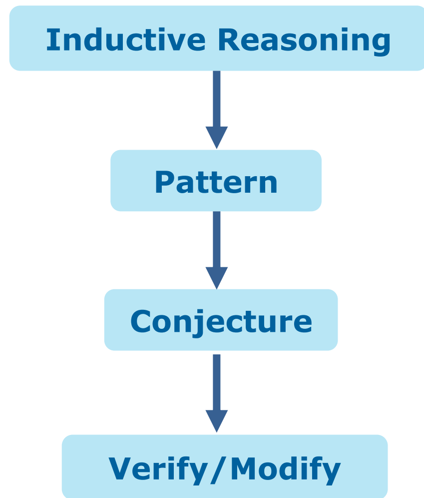
Blue	Orange	Green	Total
111	289	72	472

There are two types of reasoning that can be used to prove a conjecture.

Deductive reasoning uses facts and general principles to draw conclusions.



Inductive reasoning uses specific observations and patterns to draw conclusions.



5. Discuss with your partner which type of reasoning best describes the type that was used in the warm-up. Summarize your discussion.

Much of the reasoning that will be used in geometry is deductive reasoning.

Deductive reasoning needs to be used to prove the following statement is true.

All numbers ending in 0 or 5 are divisible by 5.

To verify that ***all*** numbers that end in 0 or 5 are divisible by 5 would require more than the method of verifying and modifying conjectures. When trying to show that something works for all cases, deductive reasoning should be used.

6. The following statement is an example of inductive reasoning.

All small dogs in the park today have blue collars.
Therefore, all small dogs have blue collars.

Discuss with your partner if it is possible that there existed a scenario that could have led to the conclusion. Write a short explanation.

A ***counterexample*** is a statement that disproves a conjecture.

7. Write a counterexample that would disprove "all small dogs have a blue collar."

2.2.3 Bring It Together: Using Reasoning

1. Determine the type of reasoning for each statement.

Statement	Type of Reasoning
The car's battery provides power to the engine, so if the battery is dead, the car won't start.	
I know Joe is a good cook because I've eaten at his house three times and each time the food has been amazing.	
The diary that is supposed to be from the 18th century might be real. It has passed a number of forensic and historical tests.	
The sum of two 2-digit numbers is always a 3-digit number.	

2. For each statement, write a counterexample that disproves the statement.

   a. If a number is multiplied by itself, then the product is even.

   b. Adding the same number to both the numerator and the denominator of a fraction does not change the fraction's value.

2.2.4 Wrap-Up: Cool Down

1. Write a statement that uses deductive reasoning.

2. Write a counterexample to the statement.